



Algorithms and Data Structures

Course Syllabus



COURSE DESCRIPTION

This course delves into the fundamental principles of Algorithms and Data Structures, exploring their significance in computer science, software development, and problem-solving.

Participants will comprehend the core concepts, algorithms, and structures that underpin efficient data management and algorithmic efficiency. From basic structures to advanced algorithms, this course empowers learners to design, implement, and analyze algorithms and data structures in various scenarios, enhancing their problem-solving prowess and computational thinking.

COURSE INFO

Prerequisite knowledge

- Basic programming skills in any language
- Understanding of fundamental mathematical concepts like algebra and calculus

Target Audience

- Computer Science undergraduates seeking a deeper understanding of algorithms and data structures
- Software engineers aiming to enhance their problem-solving abilities and algorithmic thinking
- Professionals involved in software development interested in optimizing code efficiency

Additional Information

- Emphasis on real-world applications and case studies throughout the modules.

MODULES

1. Introduction to Data Structures
2. Fundamental Algorithms
3. Intermediate Data Structures
4. Advanced Algorithms and Techniques
5. Practical Application and Performance Analysis
6. Memory Management and Caching
7. Parallel and Distributed Algorithms
8. Specialized Data Structures
9. Machine Learning and Algorithmic Techniques
10. Geometric Algorithms
11. External Memory Algorithms
12. Security Algorithms
13. Algorithmic Challenges in Big Data
14. Real-world Algorithmic Problem Solving

Note: The modules and subtopics provided here are suggestions and can be tailored based on the specific requirements.

MODULE 1

Introduction to Data Structures

- Arrays and Linked Lists
- Stacks and Queues
- Introduction to Trees and Graphs
- Time and Space Complexity Analysis
- Basic Sorting and Searching Algorithms

MODULE 2

Fundamental Algorithms

- Divide and Conquer
- Greedy Algorithms
- Dynamic Programming
- Recursion and Backtracking
- Algorithmic Paradigms and Strategies

MODULE 3

Intermediate Data Structures

- Binary Trees and Binary Search Trees
- Heaps and Priority Queues
- Hashing and Hash Tables
- Balanced Trees (AVL, Red-Black Trees)
- Graph Algorithms (Traversal, Shortest Path)

MODULE 4

Advanced Algorithms and Techniques

- Advanced Sorting Algorithms (Merge Sort, Quick Sort)
- String Algorithms (Pattern Matching, Edit Distance)
- Advanced Graph Algorithms (Minimum Spanning Trees, Network Flow)
- NP-Completeness and Approximation Algorithms
- Randomized Algorithms and Probabilistic Data Structures

MODULE 5

Practical Application and Performance Analysis

- Algorithm Design Patterns in Software Development
- Case Studies: Applying Algorithms in System Design
- Performance Metrics and Analysis Techniques
- Benchmarking and Profiling
- Optimizing Algorithms for Memory and CPU Efficiency

MODULE 6

Memory Management and Caching

- Memory Hierarchy and its Impact on Algorithms
- Cache-aware Algorithms
- Memory Allocation and Deallocation Strategies
- Optimizing for Cache Locality
- Memory Profiling and Debugging

MODULE 7

Parallel and Distributed Algorithms

- Parallel Computing Fundamentals
- Parallel Sorting and Searching
- Distributed Data Structures
- MapReduce and Parallel Processing
- Synchronization and Coordination in Distributed Systems

MODULE 8

Specialized Data Structures

- Trie and Suffix Tree
- Disjoint Set Union (Union-Find)
- Bloom Filters and Count-Min Sketch
- Self-balancing Trees (Splay Trees, Treaps)
- Spatial Data Structures (Quadtree, KD-Tree)

MODULE 9

Machine Learning and Algorithmic Techniques

- Algorithms in Feature Engineering
- Optimization Algorithms in Machine Learning
- Decision Trees and Ensemble Methods
- Clustering Algorithms
- Neural Networks and Backpropagation

MODULE 10

Geometric Algorithms

- Convex Hull Algorithms
- Closest Pair of Points
- Line Segment Intersection
- Voronoi Diagrams
- Computational Geometry in GIS

MODULE 11

External Memory Algorithms

- External Sorting
- B-trees and External Memory Search
- Multiway Merge and Polyphase Merge
- I/O-efficient Graph Algorithms
- Disk-based Hashing

MODULE 12

Security Algorithms

- Cryptographic Hash Functions
- Public Key Cryptography Algorithms
- Digital Signatures and Certificates
- Secure Multiparty Computation
- Homomorphic Encryption

MODULE 13

Algorithmic Challenges in Big Data

- Scalability and Distributed Systems
- Streaming Algorithms
- Approximate Query Processing
- Data Sketching and Sampling
- Large-scale Graph Processing

MODULE 14

Real-world Algorithmic Problem Solving

- Identifying and Analyzing a Real-world Problem
- Designing and Implementing an Efficient Algorithmic Solution
- Evaluating Performance and Optimizing
- Presenting and Justifying Design Decisions
- Peer Review and Collaboration

צרו קשר

מי אנחנו?

חברה המעבירה הדרכות והשתלמויות בעולמות הטכנולוגיה וההייטק לארגונים ששואפים גבוה ורוצים לגעת בשיא הטכנולוגיה. ההדרכות מותאמות באופן אישי לכל מטרה, כאשר תכני הלימוד נבנים בהתאם לצורכי הארגון, יעדיו ודרישותיו. CodePro משמשת גם ככוח אדם ומספקת לארגונים מרצים, מפתחים ויועצים טכנולוגים על פי דרישה.

בנוסף לתחום ההדרכה, ל- CodePro בית תוכנה ומחלקת ייעוץ המלווה חברות הייטק וגופים טכנולוגיים בתהליכי פיתוח ממשקים מאתגרים. לצוות יכולת להעניק ייעוץ ותמיכה במקרים מורכבים בהם דרושה טכנולוגיה ספציפית, כמו גם בתהליך הקמת צוותים טכנולוגים בארגונים.

נשמח לשמוע ממכם!

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